THE NOETHERIAN FIBRE FLATNESS CRITERION FOR COMPOSABLE MAPS OF SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6031-6051 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6053-6063, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (spaces, more general; both sides complete). Statement self-contained; proof is chart-reduction plus one exact citation — borderline-adjacent, kept under the written-reduction reading.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma flat-on-fibres-at-point; More on Morphisms, theorem criterion-flatness-fibre-Noetherian.

Source reference. Original theorem 05WY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5632–5662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 039B: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6063–6063. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1. *Let S be a scheme. Let $f : X \to Y$ and $Y \to Z$ be morphisms of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Assume*

(1) *X, Y, Z locally Noetherian, and*
(2) *$\mathcal{F}$ a coherent $\mathcal{O}_X$-module.*

Let $x \in |X|$ and let $y \in |Y|$ and $z \in |Z|$ be the images of x. If $\mathcal{F}_{\overline{x}} \neq 0$, then the following are equivalent:

(1) *$\mathcal{F}$ is flat over Z at x and the restriction of $\mathcal{F}$ to its fibre over z is flat at x over the fibre of Y over z, and*
(2) *Y is flat over Z at y and $\mathcal{F}$ is flat over Y at x.*

Proof. Choose a diagram as in Lemma 05WY part (3). It follows from the definitions that this reduces to the corresponding theorem for the morphisms of schemes

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$U \to V \to W$, the quasi-coherent sheaf $a^*\mathcal{F}$, and the point $u \in U$. Thus the theorem follows from the corresponding result for schemes which is More on Morphisms, Theorem 039B. $\square$